Physics-Guided Deep Neural Networks for Radar-Based UAV Recognition in Different Environments With No Prior In Situ Data

SEAN KEARNEY, Student Member, IEEE
SEVGI Z. GURBUZ, Senior Member, IEEE
North Carolina State University, Raleigh, NC USA

Classifying uncrewed aerial vehicle (UAV) radar micro-Doppler signatures with deep neural networks (DNNs) suffers a degradation in performance when there is a difference in the distribution of training and test data due to the differences in clutter backscatter and signal-to-interference-plus-noise ratio (SINR) of different environments. This is further exacerbated in low-SINR samples in which the helicopter motor rotation (HERM) line characteristic of UAV blade rotations becomes only faintly visible. As HERM lines are a critical feature enabling the recognition and discrimination of various types of UAVs, this article addresses the twofold challenge of 1) how to accurately synthesize HERM lines in training data for UAV classification and 2) how to enhance recognition performance in previously unseen different environments, especially when the SINR is low. Toward these aims, we propose a physics-guided approach in which a semantic segmentation network trained only using low-fidelity computer aided design (CAD)-models of UAVs is integrated into the discriminator of a generative adversarial network (GAN) for training data synthesis, and on the front end of a DNN for classification. Our results show that physics-guided semantic segmentation improves the representation of HERM lines in GAN-synthesized samples and enables improved classification of low-SINR samples exhibiting HERM lines when no prior in situ data at that SINR are used during training. Consequently, our approach provides a way of enhancing UAV recognition performance in different operational environments, from which it is often difficult to obtain training data prior to deployment.

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Authors' address: Sean Kearney and Sevgi Z. Gurbuz are with the Department of Electrical and Computer Engineering, North Carolina State University, Raleigh, NC 27695 USA, E-mail: (sjkearne@ncsu.edu; szgurbuz@ncsu.edu). *(Corresponding author: Sevgi Z. Gurbuz.)*

I. INTRODUCTION

Deep learning has become a powerful tool enabling advanced capabilities in modern sensing systems. In particular, deep neural networks (DNNs) have been proposed for radar-based classification [1], [2] of ground-based targets, such as humans and vehicles, as well as aerial targets, such as UAVs, larger aircraft, and even birds. In particular, utilization of the radar micro-Doppler (μD) signature has been used by many researchers for UAV detection [3] and classification [4], [5], [6]. μD signatures were also shown to be effective for classifying UAVs with or without a payload [7], [8], distinguishing UAVs from other small targets such as birds [9], [10], [11], [12], as well as for implementing multitask learning to detect characteristics of a UAV such as the number of wings and the presence of a payload [13].

However, deep learning techniques rely upon the availability of large amounts of data for training models. Unlike computer vision applications for which there are numerous publicly available repositories of imagery, there are few suitable repositories of radar data for training. Moreover, utilizing radar data acquired with a different transmit waveform than that deployed to train deep models can also result in massive performance degradations [14]. Acquisition of training data can also be costly, time consuming, and still fail to span all the different environments in which the radar will be deployed.

Thus, a critical challenge is how to train deep models so that they remain effective when the radar system is operating in a previously unseen different operational environment, from which prior acquisition of in situ training data is not possible. When a pretrained network is used to classify data in a different environment, the difference in distribution between training and test data results in significant performance degradations. In this work, we define and characterize the environment based on the signal-to-interference-plus-noise ratio (SINR). Sample distributions are impacted not only by power of the radar return from the target, but also by the clutter distribution of the environment.

A common approach to address the lack of sufficient training data is to generate synthetic training data using generative adversarial networks (GANs) [15], [16], [17]. However, because radar data are fundamentally not images, but are instead a time-varying complex data stream, radar signal processing techniques are utilized to compute 2-D or 3-D tensors, which expose the range, Doppler, and angle measurements of the radar. However, the relationships between these physical variables result in computed images whose pixels do not embody the same spatial correlations as those of optical images. Radar images will only have high pixel intensities at the range, Doppler, and angle corresponding to the target motion. Thus, many types of GANs, including Wasserstein GANs (WGANs), auxiliary conditional GANs (ACGANs), and generative style transfer networks, such as Pix2Pix and CycleGAN, have been shown to generate kinematically impossible samples when applied to radar μD signatures [18], [19], [20]. These erroneous synthetic samples cause significant losses in classification performance. Kinematic sifting [18] by identifying and discarding such erroneous samples has mitigated such losses; however, a better solution is to design GAN architectures that are guided by physics-based models to avoid generating erroneous samples in the first place. To account for target kinematics during synthesis, a physics-aware multibranch GAN (MBGAN) was proposed in [19], in which the envelope of the μD signature was then utilized both as an input to a second auxiliary branch of the discriminator, as well as to compute an additional physics-based loss term in the cost function of the GAN.

However, these aforementioned works focus on human signatures for human activity and gesture recognition, where the μD signature envelopes are bounded by the maximum velocity of the body. In contrast, the kinematics of UAV μD signatures are characterized by helicopter motor rotation (HERM) lines, which appear as lines parallel to the return of the body of the UAV. The appearance of HERM lines is dependent on the length of the window used for the STFT when generating spectrograms, with large STFT windows resulting in HERM lines [21]. Accurate representation of HERM lines in synthetic samples is critical because they are a rich source of information about the UAV, such as the number of propellers as well as each propeller's rotation rate. Conventional GAN-synthesized samples and those of the envelope-based physics-aware GAN (PhGAN) result in synthetic samples having a disjoint appearance and lack the parallelism and periodicity of real HERM lines. We show that by failing to accurately capture HERM line kinematics, such synthetic samples exhibit degraded performance when used as training data.

Alternative to GANs, physics-based methods for high fidelity synthesis [22], [23], [24] of the radar returns from a UAV have been proposed as well. However, such methods require prior knowledge of UAV dynamics for simulations, whereas target dynamics can vary between specific drones (e.g., one rotor spins faster than another) or can dynamically vary based on weather conditions (e.g., wind speed). Moreover, high-fidelity simulations are typically computational intensive and generate samples consuming a lot of memory as well. Consequently, simplified simulations of radar returns were proposed in [21], [25], and [26]. Each of these simulation methods reduces the computational cost as well as the high degree of prior knowledge needed to create more high fidelity models. Further details on each of these simulation approaches as well as their shortcomings are detailed in Section II-A.

While physics-based simulations may result in realistic representations of target kinematics, an important deficiency is that they do not include simulation of the clutter backscatter from the environment, SINR is not accounted for. To account for the environment during simulation, the authors in [27] and [28] each used real radar samples collected when no target was detected and summed the simulated complex time-series data with the extracted background data. While this method created synthetic samples with realistic environment noise, it requires a large dataset of prior real data in the same environment.

The impact of varying signal-to-noise ratio (SNR) was also studied in [29]. While classifying UAVs versus birds, this work injected noise into samples to reduce the SNR of testing data. It was found that the performance of DNNs degrades as the SNR decreases, but shallow CNN architectures are more robust to noise, and their performance decreases at a slower rate. A shallow CNN was again used in [30], where the degradation of the network based on the level of signal-to-background ratio (SBR) of UAV radar returns was studied. When introducing multiple types of drones into the drone versus bird classification, the CNN showed poor performance when testing on low-SBR data, with the smallest drones being misclassified as birds. To boost classification performance, it was found that low-SBR samples needed to be injected into the training dataset to see an improvement in over 20% accuracy. While each of these methods studies the impact of noise on the accuracy of classification with DNNs, they only study how the strength of the signal impacts classification and not the impact of distortion from the noise of the environment on classifiers.

In this article, we propose a dynamic data-driven application system (DDDAS) [31], [32] approach that aims at leveraging the strengths of physics-based synthesis and generative learning to address the two-fold challenges of 1) how to generate synthetic samples for training that accurately represent HERM lines and background clutter backscatter and 2) how to enhance classification performance in different environments without the use of prior in situ data, especially when the SINR is low. In particular, we develop a physics-guided approach in which a semantic segmentation network trained only using low-fidelity CAD-models of UAVs is 1) integrated into the discriminator of a GAN for synthesis of kinematically accurate training data and 2) exploited on the front-end of a DNN for enhanced classification in different environments. While the semantic segmentation of range–Doppler map (RDM) and range–angle map (RAM) has been investigated in [33], [34], [35], and [36], to the best of our knowledge, this article is the first to consider physics-based semantic segmentation of radar μD and how semantic segmentation can be exploited for μD synthesis and classification.

The rest of this article is organized as follows. We discuss in Section II related works on UAV μD simulation and implementations of semantic segmentation. Section III describes the dataset collected for this work, how it is processed, and then splits into further subsets based on the noise of the environments the data were originally collected in. The impact of environment on the SINR and feature space of the resulting dataset is discussed in Section IV. The method for simulating low-fidelity UAV μD signatures from CAD models is explained in Section V-A as well as synthesizing new samples with GANs in Section V-B, while Section VI-A then describes the training of the semantic segmentation network using low-fidelity physics-based simulations. Semantic segmentation results on real UAV μD signatures are also shown. Section VI-B describes the proposed Semantic-Segmentation-based GAN (SS-GAN) network for generative synthesis of kinematically accurate UAV μD signatures, while Section VI-C presents results showing how DNNs trained on this synthesized dataset yield improved classification performance on real UAV data at a similar SINR level. Section VI-D then presents results for the case of classification on previously unseen low-SINR samples. Section VII describes the proposed SS-DNN classifier, which integrates semantic segmentation into the front-end of a DNN to further improve classification in a previously unseen, different, low-SINR environment. Section VIII presents discussion and future work. Finally, Section IX concludes this article.

II. RELATED WORKS

The approaches proposed in this article relate to the synthesis of the radar return from UAVs and resulting data representations (e.g., μD and range–Doppler maps), as well as to semantic segmentation and classification in previously unseen low-SINR environments. Thus, in this section, we discuss in more detail the related works to these topics.

A. UAV Signature Synthesis

Current literature proposing methods for simulating UAV radar returns can be split between methods that result in either high- or low-fidelity simulations of the physics governing UAV dynamics and the resulting radar return. Examples of high-fidelity physics-based models of radar return from a UAV include [22] and [23]. In [22], the surface of each blade was partitioned into smaller patches, which are then each modeled as a point-scatterer. The radar return from each full blade was then computed as the sum of the backscatter from each patch, where the radar cross section (RCS) of each patch is known prior. The body of the UAV was also modeled as primitive shapes whose RCS is also prior known. The Doppler frequency was then simulated according to the drone orientation relative to the radar for the full simulation. To accurately simulate the return of the UAV, the rotor speed and orientation of the rotors throughout the flight must be specified. Therefore, this model requires a high degree of prior knowledge of the RCS of point scatterers of the blades as well as the flight dynamics of a UAV.

Likewise, a high degree of prior knowledge of UAVs was also used in [23]. To simulate the μD return, the UAV was represented as a collection of point scatterers whose position was updated according to a kinematic and dynamic model of the UAV. Specifically, the known kinematics and dynamics of a UAV in flight were used to update the changes in position and frequency of rotors according to a prior flight path of a real UAV. By taking into account the dynamics of flight, this model created realistic μD returns in a noiseless environment.

Alternatively, Sayed et al. [37] utilized a full-wave EM CAD tool to simulate the range–Doppler map of different UAVs. This simulation method required prior knowledge of the mechanical controls of the UAV to then best emulate the changes in the range–Doppler returns when the UAV is moving in particular directions. While the method showed improved results when classifying UAVs in different motions, the training and testing datasets were both simulated; therefore, the issue of clutter and low SINR incurred in real outdoor measurements was not addressed. Another method for utilizing CAD models for simulating UAVs was presented in [24]. The UAVs were scanned to create high density 3-D CAD models such that the individual propellers of the UAV were sampled initially at 30 million points before then being downsampled to a grid space of 0.3 mm. Upon simulating the radar return of the UAV, the μD was computed with a short window so that blade flashes were observed instead of HERM lines. While each of these models simulates physically accurate radar returns of UAVs, they are hindered by requiring a large prior knowledge of the dynamics and/or radar signature of the UAV being simulated, as well as being computationally costly to run.

The thin-wire model [25] is a simpler simulation approach, in which the propeller is electromagnetically modeled as a thin wire. This simulation method has proven to be effective in simulating noiseless radar returns and has been validated in such works as [25] and [26]. These works showed that the thin-wire model is able to generate rudimentary HERM lines while using a limited number of scattering points (one for each blade). Similarly, Rahman and Robertson [21] simulated the time-domain backscatter from each rotor blade as individual thin wires, finding the total returned signal according to the length of each blade, rotation rate of each blade, and the total number of blades. Although the thin-wire model enables an efficient low-fidelity simulation of UAV μD returns, the resulting μD signatures exhibit visually noticeable differences from real UAV μD signatures. Such discrepancies result in degraded classification performance when the synthetic samples are used for training.

B. Generative Synthesis of Radar Data

After the great contributions of GANs for synthesis of optical imagery for computer vision, many researchers have sought to translate these results to radar datasets. However, given that radar data are inherently a time series of complex in-phase (I) and quadrature (Q) data, radar signal processing techniques must be applied to obtain computed images that reveal range, Doppler, and angle information of a target. Commonly, radar data are represented as a 2-D RDM or RAM, which can be further formed into a 3-D tensor as a time series of 2-D images. Alternatively, 2-D μD signatures revealing the variation of Doppler frequency with time can be computed as the spectrogram—the square modulus of the STFT—of the data. Initial works on GAN-based synthesis of radar data involved application of existing architectures on μD or RDMs, showing that synthetic data could boost the efficacy of training deep convolutional neural networks (CNNs) [38], [39], autoencoders (AEs) [40], and convolutional autoencoders (CAEs) [41]. ACGANs were proposed as a variant especially effective in μD synthesis, surpassing the performance provided by WGANs or conditional variational AEs [18], [38]. In [42], conditional GANs were utilized to synthesize the passive RF spectrum obtained from humans to train a CNN or k-Nearest Neighbor classifier for occupancy detection. In [43], RDMs synthesized using a WGAN under varying degrees of occlusion were compared with simulation-based ray tracing.

As the correlations between pixels are not merely spatial but dependent upon the kinematic properties of target motion, the ability of traditional GANs for accurately representing radar target signatures is limited. While kinematic errors in GAN-synthesized μD were first demonstrated in [18], various researchers have proposed modifications to GAN architectures to improve synthesis of the spatiotemporal relations in radar data. In [44], a novel GAN-based one-class classifier for human motion detection was developed in which a dual generation scheme, consisting of an AE and generator to automatically produce new μD samples to train the discriminator on both the pixel and semantic level, was proposed. In [45], the introduction of a refinement model downstream from the generator in the GAN was proposed, which demonstrated higher stability and accuracy in generating augmented spectrograms. In [19] and [20], μD envelopes are provided as an auxiliary input to the discriminator and basis for computing a physics-based loss term to improve the kinematic fidelity of GAN-synthesized human μD signatures [19], [20]. Significantly, GANs have also been proposed to bridge the gap between simulated and real radar μD signatures [20], [46] and differences in the radar transmit frequency used in training and testing data [14], as well as synthesize target signatures at different aspect angles than that of the trainingbrk data [47].

While the aforementioned works take as input only one type of 2D image, either μD or RDM, RDGan [48] takes as input a radar data stream and computes a sequence of RDMs using a two-discriminator architecture inspired by TempoGAN [49]. TempoGAN is a GAN architecture originally desired to simulate time-varying fluid flow, which takes as input a time series of images and utilizes a spatial discriminator and temporal discriminator to determine whether synthesized 3-D flow is real or fake. RDGan employs a similar structure, but additionally computes time–Doppler spectrograms from the synthesized RDMs to learn an enhanced set of statistical and time–frequency features using a multiscale CNN. The approach is shown to give improved recognition accuracy of five different pedestrians with a 77-GHz automotive radar.

While there is an abundance of work considering the synthesis of radar returns from human activities or gestures, there is very little work on synthesis of UAV signatures. A notable exception is a recent work [50] in which RDMs were synthesized using a GAN for data augmentation, which was shown to boost the classification performance of a CNN trained to recognize three-types of four-wing drones of different sizes. Thus, to the best of our knowledge, this article fills an important gap in the literature on UAV signature synthesis, developing physics-aware methods for UAV synthesis and classification that offer greater kinematic fidelity and accuracy under low-SINR scenarios.

C. Semantic Segmentation of Radar Data

This article proposes physics-guided semantic segmentation (PhG-SS) as a way of capturing and informing DNNs of the kinematic properties of UAV flight, thus ensuring the accuracy of HERM line synthesis. The current literature on semantic segmentation of radar data is limited and focused on applications of synthetic aperture radar (SAR) image segmentation or automotive radar for object recognition. For example, in [51], a semantic segmentation network for point cloud data, PointNet, was created which showed an ability to directly segment objects from point cloud data. Another implementation of semantic segmentation on point cloud data, in this instance collected with a frequency modulated continuous wave (FMCW) radar, is demonstrated in [33]. To implement semantic segmentation, a fully connected CNN (RSS-Net) was created and trained using information from LiDAR and RGB data to better improve the ability of the network to segment objects in radar point cloud data. Other implementations of semantic segmentation on FMCW radar data where the Doppler information was utilized can be seen in [34], [35], [36], and [52]. In [52], a multi-input multi-output encoder–decoder network was created to take as input RAM, RDM, and angle–Doppler map to then segment the location of objects in the range–angle and range–Doppler views. Using RSS-Net as a multi-input multi-output network, Zhang et al. [35] made improvements on RSS-Net by updating the encoder for RDMs to now include a new convolutional operation: PeakConv. PeakConv takes inspiration from constant false alarm rate (CFAR) in classical radar signal processing and showed improvements in segmentation over RSS-Net. A novel semantic segmentation model, fast Fourier Transform (FFT)-RadNet, was created in [34], where the input consists of only RDMs, and the network has multitask heads for both object detection and segmentation. Finally, Dalbah et al. [36] introduced an attention block in a novel segmentation architecture: TransRadar, a multi-input network for radar segmentation. By introducing an adaptive-directional attention block, TransRadar showed improved segmentation of RDM and RAM in comparison with FFT-RadNet.

Each of these works shows a use of semantic segmentation with radar Doppler information, not radar μD. Therefore, it is to the best of our knowledge that the problem of semantic segmentation of μD signatures, and, in particular, UAV μD signatures, has not been considered in the literature to-date.

D. Classification in Disparate or Low-SINR Environments

The third major component of this article is the ability of a deep learning model to classify radar data collected in disparate or low-SINR environments. When classifying using radar μD data, the noise from the background environment can significantly occlude the target return. Prior work has studied through-the-wall radar imaging, where the wall acts as a source of noise that similarly occludes the target. In [53], an autofocusing technique based on higher order statistics was presented and shown to create high-quality radar images through walls without prior known information on its characteristics. This technique showed an ability to clearly focus on stationary targets seen through-the-wall. The use of FMCW radar for through-the-wall classification of human μD was explored in [9]. To classify human μD through-the-wall, a radar was set 0.5 m outside residential buildings, and a singular value decomposition was used to classify whether a person was walking with an object in hand or not. While these works show an ability to detect through-the-wall using radar, the noise created by a wall is distinct from that of an outdoor environment. Taking into account the environment, Zenaldin and Narayanan [54] explored utilizing machine learning to classify radar μD of human activity in both indoor and outdoor environments. In this work, a support vector machine (SVM) was trained to classify three different motions on μD data collected in both indoor and outdoor environments, with the SVM achieving a classification accuracy over 80%. The impact of environmental noise on the μD signatures from human activity though is significantly reduced when compared to that on UAVs, as the return power from a human target is significantly stronger than that of a small UAV.

Other uses of radar where the noise caused by the environment is considered are found in SAR imaging applications. Speckle is a common source of noise in SAR imaging; therefore, Wang et al. [55] explored the ability to use deep learning for classifying targets in SAR images at varying speckle levels. A dual-stage CNN was created in this work, with the first stage despeckling the input SAR image and then the second stage classifying targets. This network showed classification accuracy higher than 82% at a variety of speckling levels. Another method to improve the classification of deep learning models is to use synthetic data for training. In [56], only synthetic data were used for training a deep learning automatic target recognition (ATR) model. The dataset used in this work was the publicly available SAMPLE dataset from U.S. air force research laboratory (AFRL). This work showed that training with only synthetic data resulted in significant challenges when generalizing to real data. To bridge this gap, the work showed that use of data augmentation, a residual network with dropout layers, as well as the choice of the model loss function can improve the ability of the model, with the best model presented achieving over 95% testing accuracy on real data. While this work did not address the challenge presented by noisy environments, it did show that deep learning can be implemented to classify data from disparate domains, which in this work was synthetic versus real data.

Each of these works considers the classification of radar data in either noisy environments (such as through-the-wall) or when the testing dataset is disjoint from the training dataset. In [57], this scenario was specifically studied for UAV radar μD classification by varying the SNR of the spectrograms and evaluating how the discrepancies between training and testing degraded performance. Noise injection was utilized to equalize the SNR of training and test data, and improved the classification of UAVs and birds from 81% to greater than 90%. The classification of UAV spectrograms at different SNR levels was also studied in [58]. Unlike other drone classification works discussed, the spectrograms in this dataset were created using the RF signal from the drone's controller instead of the μD return. While this work did show that a CNN model could be effective when trained on samples with different SNR levels than that being tested, the measure of SNR was according to the noise observed in the controller's transmit signal, not from external environment created noise that obscures the μD return of the UAV. Therefore, to the best of the authors' knowledge, prior works have not adequately considered the problem of how to maintain classification performance when dealing with previously unseen (no prior in situ data) data at low SINR.

TABLE I
Data Collection Radar Parameters

Radar Parameters	
Center Frequency	79 GHz
Bandwidth	4 GHz
PRF	6.4 kHz
Radar Height	1.5 m
Operating Time	30 s

TABLE II
UAV Characteristics

UAV	Propeller Length	Span
Small Hexacopter	230 mm	0.86 m
Large Hexacopter	230 mm	1.31 m
Small Quadcopter	147 mm	0.24 m
Medium Quadcopter	230 mm	0.55 m
Large Quadcopter	230 mm	1.4 m

III. UAV DATA AND PROCESSING

A. UAV Data Collection

For the purpose of this article, radar data were collected on a variety of UAVs using the TI AWR2243 single-chip 77- to 81-GHz radar. This FMCW radar was operated with a waveform utilizing the full 4-GHz bandwidth and a pulse repetition frequency (PRF) of 6.4 kHz, as defined in Table I. The radar was placed 1.5 m above the ground and operated for increments of 30 s per sample. The UAVs contained in this dataset are described in Section III-A, and then, the method for processing the dataset for use in deep learning is explained in Section III-B.

Utilizing the FMCW radar, data were collected on five UAVs, which varied in terms of size, number of propellers, and length of propellers. Each of the UAVs is displayed in Fig. 1 and described in Table II, where it can be seen that the target set consists of: two Hexacopters and three Quadcopters. The smaller of the two hexacopters has a propeller length of 230 mm and a span of 0.86 m, while the larger hexacopter has a propeller length of 230 mm and a span of 1.31 m. Among the quadcopters, the propeller lengths utilized were 147, 230, and 230 mm, which each had a corresponding span of 0.24, 0.55, and 1.4 m, respectively.

(a) (b) (c) (d) (e)

Fig. 1. UAV data collected on (a) small, (b) medium, and (c) large Quadcopters as well as (d) small and (e) large Hexacopters

TABLE III
Data Collection UAV Parameters

UAV Models	Environments	Flight Pattern	Air Temperature	Wind Speed
Small Hexacopter Medium Quadcopter	Open Field Car 10m away Tree 10m away	Hover	24°C	$\leq$ 8kph
Large Hexacopter Small Quadcopter Large Quadcopter	Open Field	Hover	33°C	$\leq$ 8kph

(a) (b)

Fig. 2. (a) and (b) Operating UAVs during data collection.

These five UAVs were chosen to create a rich dataset whose radar returns should be uniquely distinguishable but will provide some overlap in distribution among the UAVs with the same number of propellers.

This dataset was collected over two different collection periods to increase the breadth of UAVs utilized for this work, as defined in Table III. The first set of data was collected on the small hexacopter and the medium-sized quadcopter. Each of these UAVs was flown around 5 m from the radar while hovering in an open field as well as with a car and then a tree [as shown in Fig. 2(a)] each 5 m behind the UAV and therefore 10 m from the radar to increase the variability in environmental noise in our dataset. Ten samples per a background environment were collected on each of these UAVs. During this collection, the weather was generally fair with a temperature around 24° C and a wind speed of no larger than 8 kph. During the second collection, the large hexacopter as well as the smaller and larger quadcopters was utilized. As shown in Fig. 2(b), each of the UAVs was flown around 5 m from the radar while hovering in an open field with eight samples collected for each quadcopter and three samples for the hexacopter given battery time constraints when flying. For this collection, the weather was hotter with a temperature around 33° C and a similar wind speed as the other data collection of no larger than 8 kph.

B. UAV μD Signatures

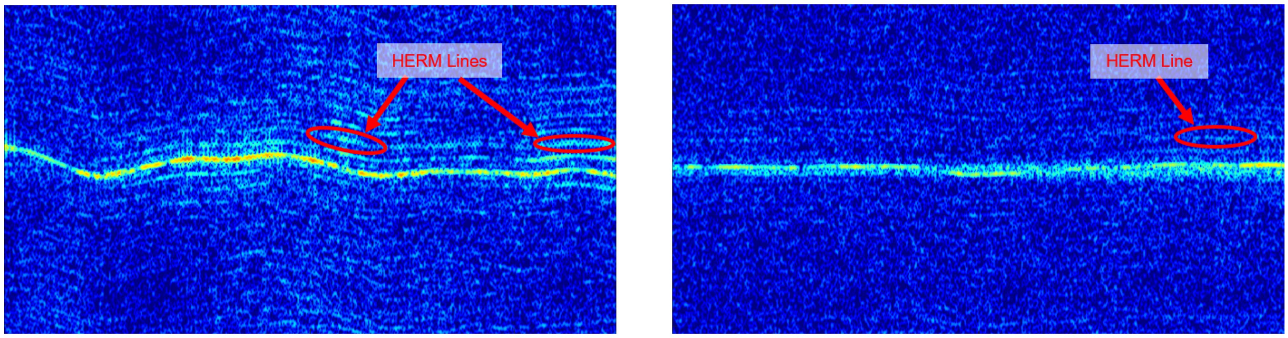
The μD signature of the UAV is computed by applying a time–frequency transform, such as the STFT, across the slow-time samples of the received radar data. Vibrational or rotational motion of an object, also known as micro-motions, results in μD frequency modulations [59], which are unique to different targets and their motions. In this work, we compute the spectrogram, or square modulus of the STFT, to represent μD signatures

$$S(m, w) = \sum_{n=0}^{N-1} x[n]w[n-m]e^{-i\omega n} \tag{1}$$

where $x[n]$ is the input signal, $w[n]$ is a Hanning window of length L, and m is the step size of the window, which is the difference between the length of the window and the overlap between windows.

When generating spectrograms of UAV data, (1) is computed in discrete time using an FFT. In this work, $N = 4096$ FFT points, a long window of length $L = 256$, and an overlap of 200 (step size of $m = 56$) were used. The length of the window size impacts whether blade flashes or HERM lines of the UAV propellers appear in the μD signature. To observe blade flashes, the rotation rate of the propellers must be lower than the unambiguous sampling rate of the radar, which, for this work, is the PRF of 6.4 kHz. Therefore, blade flashes will only be apparent for a max propeller speed of 65.8 m/s; however, the UAVs used in this work exhibit propeller speeds of around 100–150 m/s. This greatly drives up the PRF required to see blade flashes. For UAV recognition with a low PRF radar, we instead observe HERM lines by utilizing a long window in the STFT. By allowing the STFT to integrate over several rotations of the UAV blades, greater frequency resolution is obtained.

An example of the HERM lines from the smaller Hexacopter can be seen in Fig. 3(a), where the high-SINR



(a) (b)

Fig. 3. Hexacopter spectrograms: (a) High SINR and (b) Low SINR with HERM lines.

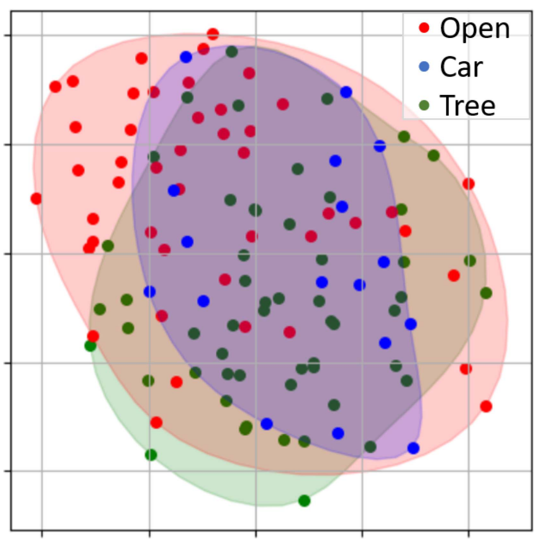


Fig. 4. Comparison of PCA features obtained across different environments from small quadcopter for data with 2–3 dB SINR.

spectrogram has clearly observable HERM lines, which are parallel to the main return and appear at consistent gaps in frequency. The frequency interval between HERM lines in the μD spectrogram directly relates to the rotation rate of the propellers of the UAV; therefore, UAVs can be distinguished by the appearance of HERM lines.

A total of 79 recordings, each 30 s in duration, across all five UAVs were obtained. These recordings were then split into shorter 6-s samples for training and evaluation, resulting in a total of 395 samples.

IV. Analyzing Impact of Different Environments

While the UAV spectrograms do typically contain noticeable HERM lines, the clutter in the background environment can obscure the target signature. Thus, radar samples were collected on UAVs in a variety of environments to vary the sources of clutter across samples. But as can be seen in the distribution of principle components shown in Fig. 4, at a specific SINR level, the distribution for different scenes was not significantly different. Therefore, we defined

TABLE IV
Distribution of SINR in the UAV Dataset

	With HERM Lines	No HERM Lines
High SINR	107	0
Low SINR	28	198

the environment not based on physical scene but instead according to the SINR levels of the μD signatures

$$\mathrm{SINR} = \frac{P_{\mathrm{signal}}}{P_{\mathrm{interference}}}. \tag{2}$$

In calculating the SINR, the signal power P_{signal} includes the power of the main radar return as well as that of the HERM lines, and the interference power $P_{\mathrm{interference}}$ is the combined power of all nonsignal components appearing in the return.

From calculating the SINR, all samples were initially partitioned into "High SINR" and "Low SINR" groups with SINRs of >10 dB and between 0 and 10 dB, respectively. Each level of SINR could then be further categorized as either having HERM lines or not. Note that samples at High SINR that did not exhibit HERM lines were pruned from the dataset, reducing the total samples to 333. The distribution of samples across SINR levels with or without HERM lines is further shown in Table IV, where there are 107 High SINR samples which contained noticeable HERM lines, 198 Low SINR samples with no visible HERM lines, and 28 Low SINR samples with noticeable HERM lines. The distribution of these samples based on principle component analysis (PCA) is then shown in Fig. 5, where the High SINR (red) and Low SINR (blue) UAV samples show distinct clustering. It may be observed that though the UAV samples with Low SINR and HERM lines (light blue) have some overlap with High-SINR samples, the centroids of each cluster are distinguishable.

To further demonstrate the difference in data at Low versus High SINR, Fig. 3 displays the spectrograms for the smaller sized Hexacopter. It can be observed that both spectrograms contain HERM lines, but the high-SINR sample has significantly more distinguishable HERM lines than the low-SINR sample. Indeed, the lack of observable HERM lines makes the classification of Low-SINR samples significantly more challenging. Hence, we evaluate our proposed

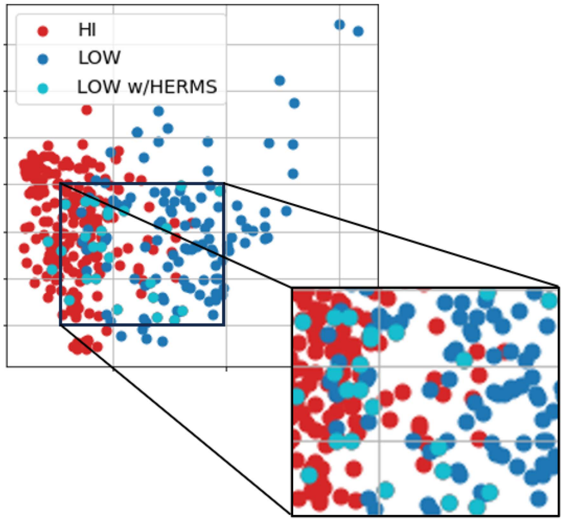


Fig. 5. PCA distribution of real data at high and low SINR.

approach by distinguishing the performance on samples with and without HERM lines.

V. UAV μD SIGNATURE SYNTHESIS

Deep learning relies on the availability of massive amounts of data to learn a model for the underlying relationships reflected in the data. However, in sensing systems, training data can be difficult and costly to obtain; moreover, the dynamic nature of the sensing environment ensures that there will always be some part of the signal that is unknown and mismatch with training data. At the heart of the DDDAS paradigm is the concept of leveraging data-driven methods as a powerful tool when used in tandem with physics-based models. The resulting hybrid approach, physics-aware machine learning, combines the strengths of deep learning and physics-based modeling to optimize tradeoffs between prior versus new knowledge, models versus data, uncertainty, complexity, and computation time, for greater accuracy and robustness. Thus, this work utilizes low-fidelity physics-based simulations to guide the process of both synthesizing training data and classifying UAV μD signatures, achieving improved accuracy in recognition of UAVs even in low-SINR conditions without requiring any prior in situ data during training. The next subsections detail our proposed approach for physics-guided generative drone signature synthesis.

A. Physics-Based Simulations

First, we use a low-fidelity physics-based model to capture the fundamental physics of blade rotations, which results in HERM lines. CAD models are used to reflect the basic physical structure of the quadcopters and hexacopters of the dataset, as shown in Fig. 6. These point clouds contain three layers: the full UAV structure, the full set of the UAV's propellers, and finally each individual propeller. In MATLAB, the layers were mapped, so that the individual propellers had a mapping to the full set of propellers, which then further had a mapping to the original drone point cloud.

Fig. 6. CAD Generated point clouds for (left) hexacopter and (right) quadcopter.

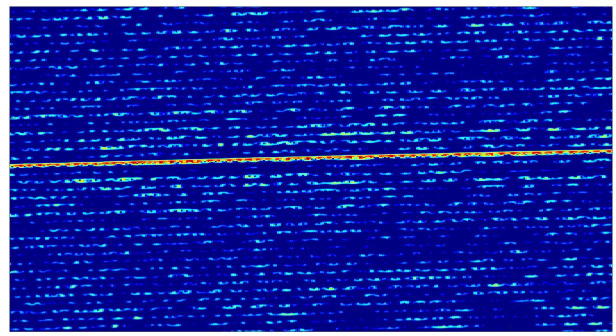


Fig. 7. CAD-model synthesized μD signature.

Utilizing this mapping scheme, ten points were sampled from each propeller as well as the body to reduce the complexity of the simulation. The remaining points were then normalized before being increased to sizes corresponding to the three quadcopters and two hexacopters, respectively.

To then simulate the motion of the UAVs, it was determined how many degrees of rotation each propeller would rotate according to its propellers' RPMs as well as how far the UAV would move per the radar's sample rate. With these values known, the UAV simulation was updated per sample by first rotating each propeller accordingly, then mapping that rotation back to the full UAV point cloud, and finally moving the full UAV to the proper position according to a predetermined randomly generated circular path. Upon generating the point cloud for each step in time for 30 s of simulation, the complex radar I/Q samples were found by calculating the change in the phase of the transmitted signal compared to the returned signal. The complex exponential was taken on this change in phase to create the complex I/Q radar return. The complex I/Q data were then processed in the same manner as the real UAV data to generate the spectrogram per simulation.

Fig. 7 demonstrates a spectrogram generated for the simulated smaller Hexacopter. It can be observed that the low-fidelity model does not provide for a like-for-like recreation of a real UAV spectrogam as seen in Fig. 3, but the simulated spectrogram does demonstrate a similar main return from the UAV as well as resulting HERM lines which are parallel to the aforementioned main return and appear at a consistent gap between one another.

When utilizing this data for training the deep learning networks discussed in Section VI-D and testing on the Low-SINR samples, it was observed that training with the low-fidelity models either with or without injected noise did not result in networks, which could classify the test set beyond random guessing. However, these simulations do embody important physics-based information about blade rotations, which we will exploit as part of our proposed approach.

B. Synthesis With GANs

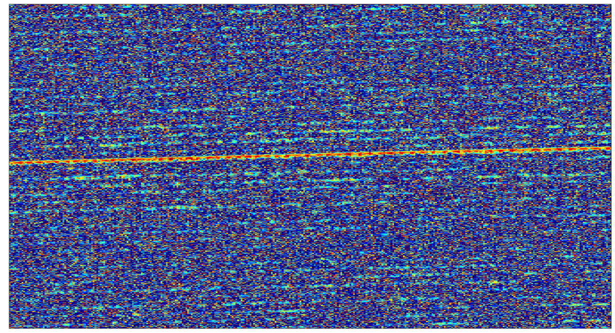
GANs have been shown to accurately synthesize images from a small dataset of original samples in many computer vision applications. These networks are able to synthesize realistic images by using a two network architecture composed of a generator and a discriminator. During training of a GAN, the generator learns to synthesize images while the discriminator tries to discriminate from real images. GAN-based synthesis has been utilized on a variety of 2-D radar imagery, including SAR, range–Doppler, and human μD signatures. However, GAN-synthesized signatures often exhibit errors in kinematics due to conventional GANs, such as the WGAN, having cost functions that only consider statistical differences. To prevent the generation of kinematically flawed μD signatures, utilization of an additional branch in the discriminator that takes as input μD envelopes and a physics-based loss term in the cost function has been proposed [19]. In particular, an MBGAN built with a WGAN as the model backbone was implemented, while dynamic time warping was used as a metric for comparing the match between synthesized and real μD signature envelopes and provided as an additional loss term. Synthetic μD signatures that captured the periodicities of gait in human ambulation with much greater accuracy were obtained. A study [19] comparing the kinematic errors incurred in synthesis of sign language signatures showed much fewer kinematic errors in comparison to WGAN synthesized signatures. We found, however, that emphasis on consistency with μD envelopes is actually detrimental to the resulting HERM lines in synthesized UAV μD signatures (see Section IV-C). Consequently, we propose a physics-guided approach that utilizes semantic segmentation to better inform the GAN discriminator about the kinematic fidelity of synthesized HERM lines. Later, we also exploit semantic segmentation for the development of an improved DNN for classification.

VI. PROPOSED APPROACH: SS-GAN

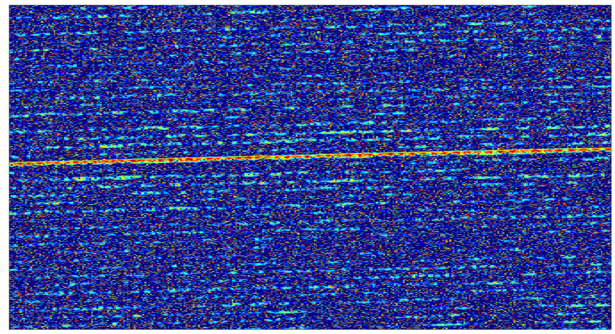
As demonstrated in Section V, current methods for synthesizing spectrograms of UAVs contain drawbacks with regard to having physically accurate μD signatures of UAVs versus realistic synthesis of background clutter. To mitigate such flaws, we develop a method that takes advantage of CAD-based UAV simulations to inform the discriminator of the expected UAV signature characteristics during synthesis. In particular, this is done by developing a PhG-SS network discussed next.

A. Physics-Guided Semantic Segmentation

To best preserve the HERM lines inherent in UAV signatures, this article proposes utilizing a semantic segmentation network to retain the information of HERM lines and distinguish it from clutter returns. We utilized the U-Net network, which has been developed for semantic



(a)



(b)

Fig. 8. Low-fidelity simulated spectrograms derived from CAD models with noise injection. (a) 5-dB SINR. (b) 20-dB SINR.

segmentation of optical imagery [60], as a backbone. The structure of the network is designed with a downsampling half and an upsampling half, which mirror one another. The downsampling half of the network contains four 2-D convolutional blocks, which not only output to the following layer, but also copy the feature map from each layer to the corresponding layers of the upsampling half of the network, allowing for information to propagate to higher resolution layers. The upsampling half is then composed of four 2-D transposed convolutional layers, which are each followed by 2-D convolutional blocks.

For this network, the loss was set according to

$$\text{Loss} = \lambda * \text{BCE} + \lambda * \text{Dice} \tag{3}$$

where BCE is the binary cross entropy, λ denotes weighting coefficients, and Dice is the dice loss, which is found by

$$\text{Dice} = \frac{2\ * \text{Area of Overlap}}{\text{Total Area}}. \tag{4}$$

The PhG-SS model is obtained by training U-Net with low-fidelity physics-based simulated UAV spectrograms from the CAD models described in Section V-A. First, the simulated UAV samples were generated with no injected noise. Then, copies of these samples were generated with White Gaussian noise injected to correspond to SINRs of 5, 10, 15, and 20 db.

Fig. 8 shows samples of the noise-injected spectrograms. Utilizing the noiseless samples, segmentation masks, shown in Fig. 9, were created from the original sample, where any pixel with signal was set to 1, and any pixel without signal was set to 0 to create a black and white

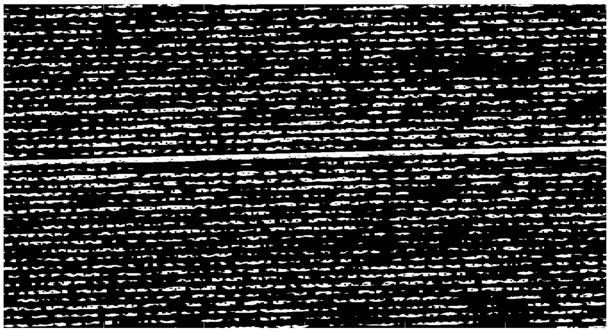


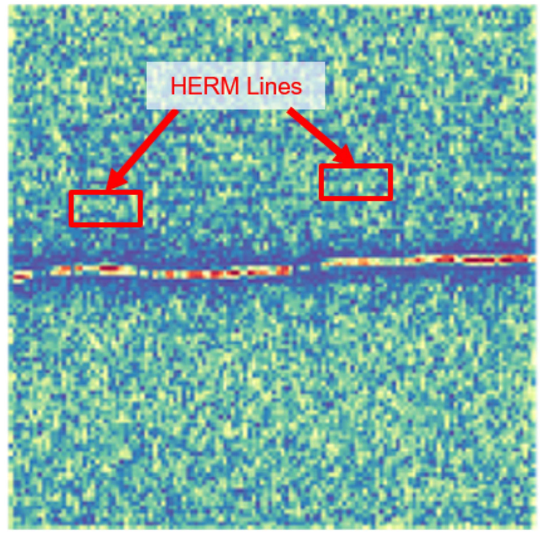
Fig. 9. SM from CAD-simulated data.

image of the noiseless signal. These segmentation maps (SMs) are then used to train the PhG-SS to recreate the corresponding SMs for each input μD signature regardless of the injected noise level.

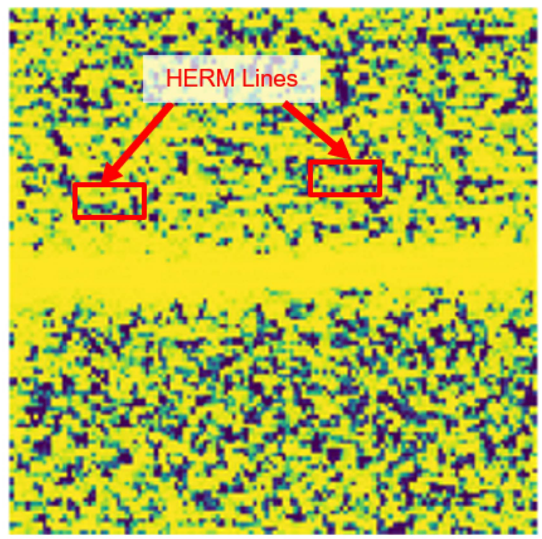
Two commonly used metrics for evaluating the efficacy of semantic segmentation of optical images are the intersection over union (IOU) score and the Dice score. The IOU evaluates the degree of overlap between a predicted (A) and ground truth (B) bounding box by finding the ratio of the area of overlap to the area of their union: $|A \cap B|/|A \cup B|$. Similarly, the Dice score provides a similarity metric by computing $2|A \cap B|/|A| + |B|$. Thus, both metrics penalize false positives. We evaluate the efficacy of PhG-SS using simulated data with various levels of injected noise, for which we obtained an IOU score of 0.388 and a Dice score of 0.559. These values are comparable to those reported in the literature for semantic segmentation of radar data (e.g., RAM and RDM) [36], [52].

However, given the appearance of HERM lines as thin curves that extend for the duration of the μD signature, metrics such as IOU and Dice, which focus on bounding boxes, may not be the best way to consider the efficacy of the PhG-SS segmentation. In this work, we consider segmentation at the pixel-by-pixel level, which is a much finer scale than bounding boxes. Moreover, metrics, such as IOU and Dice, fail to consider that in UAV μD signatures, we are more concerned with ensuring kinematic fidelity of periodicities and parallelism in the HERM lines, which may not fall neatly into bounding boxes. Thus, we consider metrics of kinematic fidelity in Section V-C, when evaluating the quality of synthetic μD signatures generated by the proposed SS-GAN, which exploits semantic segmentation with PhG-SS in its discriminator.

A visual example of an SM obtained using PhG-SS applied to a real radar UAV μD signature is given in Fig. 10. In this result, the spectrogram was resized to $128 \times 128 \times 3$ and was plotted in python with *matplotlib*'s `Spectral` colormap. The plot on the top is the original spectrogram where any pixel that contains a warm color is signal and any pixel that contains a cool color is noise. The plot on the bottom is of the same spectrogram, but with each pixel filled in with yellow that corresponds to pixels that the SM sets to 1. It can be observed that for a real sample, the HERM lines are generally mapped as being part of the SM in addition to



(a)



(b)

Fig. 10. Semantic segmentation with PhG-SS network. (a) Real μD signature at the high SINR. (b) Resulting SM.

any of the returned signal from the UAV body in the center of the spectrogram.

Thus, training U-Net with even a very rudimentary physics-based simulation can create a segmentation map that retains the information of the UAV body and the propellers' corresponding HERM lines, while ignoring background noise at a similar power level. This is a key result that we will utilize to design both a GAN with improved kinematic fidelity in synthesizing HERM lines, and a classifier that exploits semantic segmentation for denoising.

B. SS-GAN: Semantic-Segmentation-Based GAN

In our proposed SS-GAN, the PhG-SS network was added as the first stage of a new auxiliary branch in the discriminator of a WGAN with Gradient Penalty (WGAN-GP). This architecture is shown in Fig. 11, where the μD signature output by the generator is input to the main brain of the discriminator, which utilizes the μD discriminator to

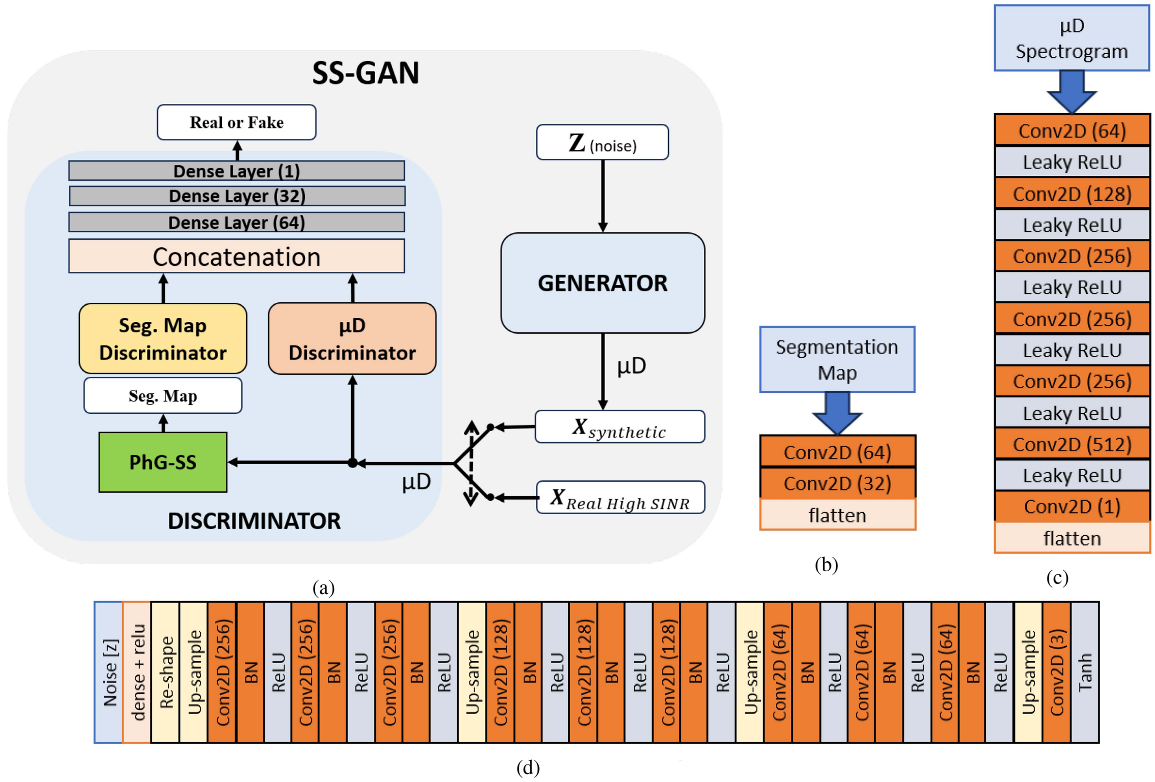


Fig. 11. Proposed Semantic Segmentation GAN. SS-GAN. (a) SS-GAN. (b) SM Discriminator. (c) μD Discriminator. (d) Generator DNN.

extract relevant μD features. In the new auxiliary branch, the generated μD signature is first input into the PhG-SS to extract an SM. The SM discriminator consists of two convolutional layers, which computes features that are specifically focused on the HERM lines. The outputs of the main (μD) and auxiliary (SM) discriminators are then concatenated and provided to the fully connected layer. Thus, with our proposed network design, the discriminator is now able to factor in the inherent physics of the HERM lines when discriminating synthesized samples. The loss function for the SS-GAN is

$$\text{Loss} = \{D(x) - D(G(z))\} + \text{GP} \tag{5}$$

where $D(x)$ is the output of the discriminator for a real input image, $D(G(z))$ is the output of the discriminator for a generated sample, and GP is the gradient penalty.

In this work, SS-GAN was trained on the μD signatures of real UAV data with high SINR. To enable performance comparisons, we also generated synthetic datasets using networks in the literature, more specifically, WGAN-GP and MBGAN [19] in a similar fashion. Each network was trained for approximately 25 000 iterations and then generated 500 samples per UAV class. These datasets were then used for training deep learning models used in the comparisons of Sections VI-C and VI-D.

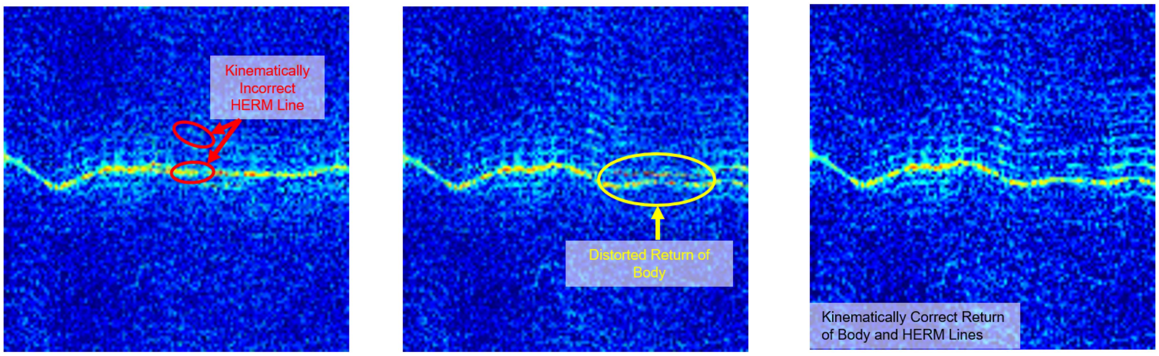
C. Kinematic Fidelity

When generating synthetic μD signatures, it is important to accurately represent target kinematics. To illustrate how current GANs can generate kinematically flawed samples, Fig. 12 shows synthetic samples of the small hexacopter's μD as generated via three different networks. It may be observed that each GAN generated a similar μD main return of the UAV, but there are distinct differences in the synthesized HERM lines and returns from the UAV body. The WGAN is able to synthesize strong HERM lines; however, as can be observed in Fig. 12(a), when there is a change in the μD shift of the UAV body, the synthesized HERM lines do not follow (stay parallel to) the corresponding shift—a kinematic flaw. Similarly, the synthetic μD signature generated using MBGAN, shown in Fig. 12(b), possesses significant distortion in its representation of the UAV body μD return. In contrast, the sample synthesized using the proposed SS-GAN, shown in Fig. 12(c), appears to maintain a strong main return from the UAV as well as kinematically correct and distinguishable HERM lines.

To quantitatively evaluate the kinematic fidelity of the HERM lines represented in the μD signatures generated, we defined a metric reflecting the fidelity of representing propeller blade rotations. More specifically, the standard deviation of the rotation rate of the propeller (σ_{ω_p}) is defined as a kinematic fidelity metric and computed as

$$\sigma_{\omega_p} = \sqrt{\frac{1}{N-1}\sum_{n=1}^{N} |\omega_{p,n} - \mu_{\omega_p}|^2} \tag{6}$$

where ω_p is the rotational rate of the propeller, N is the total number of samples per a spectrogram, and μ_{ω_p} is the average rotational rate of the propeller per a spectrogram. The rotational rate of the propeller ω_p can be found as the



(a) (b) (c)

Fig. 12. GAN synthesized quadcopter spectrograms. (a) WGAN. (b) MBGAN. (c) SS-GAN.

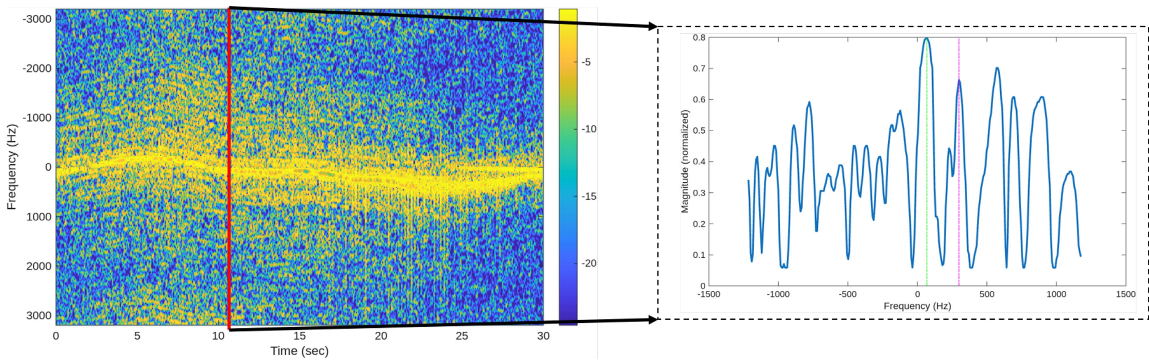


Fig. 13. Computation of propeller rotation rate from (left) hexacopter μD signature by taking a (right) slice of the μD at around 10 s, with the main body return (green line) and chopping line (magenta line) highlighted.

TABLE V
Kinematic Metrics of High-SINR Real and Synthetic UAV μD Signatures

UAV	Kinematic Metric	Dataset			
		Real @ High SINR	SS-GAN — High SINR	WGAN — High SINR	MBGAN — High SINR
Quadcopter Small	σ_{ω_p}	20.32	25.58	25.19	25.53
Quadcopter Medium	σ_{ω_p}	18.96	25.33	25.81	25.44
Quadcopter Large	σ_{ω_p}	18.44	25.87	26.10	26.27
Hexacopter Small	σ_{ω_p}	18.23	25.29	25.59	26.01
Hexacopter Large	σ_{ω_p}	18.19	25.85	26.03	26.35

frequency difference between the return of the main body of the UAV from the chopping line, which is the HERM line closest to the body response. The body response and chopping line can be identified by taking a cross section of the μD signature across frequency at a particular time, as shown in Fig. 13. The main body response corresponds to the peak of the cross section (shown with a red dashed line), while the chopping line corresponds to the first peak distinct from the body response (shown with a yellow dashed line). The propeller rotation rate is computed at each time bin within the μD signature to support computation of the standard deviation given in (6). The lower the value of σ_{ω_p}, the more consistent the representation of propeller rotation.

The results of this metric per UAV across the three GANs is displayed in Table V. The results for the real μD data at high SINR are included as a comparison for the synthesized samples to what was observed in real data. In addition, in the table, the GAN synthesized dataset with the lowest standard deviation per metric and UAV is highlighted in green, while the next best performing is highlighted in yellow. It can be observed that for four out of the five UAVs, the SS-GAN synthesized dataset had the lowest standard deviation, while the WGAN generated dataset consistently also outperformed the MBGAN generated dataset. This confirms the visual observations made on each dataset that the SS-GAN generated dataset contains less kinematic errors when synthesizing HERM lines.

Another way of assessing the differences between synthetic and real samples is to utilize t-SNE to visualize the distribution of features. For example, Fig. 14 shows the feature space for the largest quadcopter. Here, the real samples appear as green dots, while the blue dots are samples synthesized by the SS-GAN. It may be observed that around the real samples, the SS-GAN synthesized samples appear to be distributed closer to the real samples than either the WGAN or MBGAN synthesized samples. An example of one of the clusters around a real sample is highlighted on the right side of Fig. 14, where it is noticeable how the real sample circled in green is completely enclosed by SS-GAN samples, which are circled in blue.

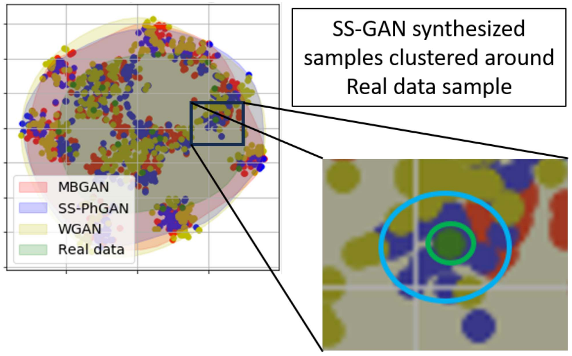


Fig. 14. Comparison of t-SNE distribution of synthetic versus real μD signatures for the large Quadcopter.

TABLE VI
DNN Classification Accuracy When Training and Testing Data From Same Environment

Network	Training Data	Testing Data	Accuracy
CNN	Real @ High SINR	Real @ High SINR	66.6%
	WGAN — High SINR		75.0%
	MBGAN — High SINR		72.6%
	SS-GAN — High SINR		76.2%
CAE	Real @ High SINR	Real @ High SINR	69.1%
	WGAN — High SINR		86.9%
	MBGAN — High SINR		80.9%
	SS-GAN — High SINR		88.7%

Next, the synthesized datasets were then compared by training two different DNNs for classification: a CNN and a CAE. The CNN is comprised of three 2-D convolution layers of 32, 64, and 128 filters, respectively. Both of the first two convolutional layers are followed by batch normalization layers, while the final convolutional layer is followed by a flatten layer and then a dense layer for classification. The CAE is comprised of an encoder network that first downsamples the input image, before input into the decoder network that upsamples to reproduce the original input image. The architecture of the encoder network consists of four convolutional blocks, where each convolutional block includes two 2-D convolutional layers followed by a rectified linear unit (ReLU) activation function before concatenation and 2-D maxpooling. The decoder network mirrors the encoder network in that it also consists of four convolutional blocks; however, instead of maxpooling the concatenated convolutional outputs, 2-D upsampling is applied. In addition, the final upsampling layer of the decoder network is followed by a 2-D convolutional layer with sigmoid activation. After training the full CAE, the decoder is removed, and layers for classification are appended to the pretrained encoder. The appended classification layers consist of a batch normalization layer, two dense layers which are each followed by a dropout layer, and finally a dense layer with softmax activation.

The results of these two networks when testing on real UAV data with High SINR can be seen in Table VI. Both networks were tested with only High-SINR data since these data contain the strongest most distinguishable HERM lines. In the table, the network architecture is specified along with the dataset utilized for training and testing the network as well as the average accuracy across five runs of each network combination. For each network, the real UAV dataset at High SINR was initially split 60/40 for training and testing, with the training dataset also being the training dataset used for each GAN and the testing dataset being the same for all instances displayed. The Synthetic Minority Over-sampling TEchnique (SMOTE) [61] was utilized to increase the number of samples used to train the CNN and CAE. SMOTE operates directly in the feature space by generating synthetic examples along line segments connecting existing minority class samples to their nearest neighbors, thereby creating new plausible samples and mitigating class imbalance and overfitting. As may be observed, both networks showed the poorest testing accuracy when trained with the real dataset at High SINR. When training the networks with GAN synthesized data, the use of the MBGAN synthesized data for training showed the poorest testing accuracy, with improvements of about 2.4% and 6% for the CNN and CAE trained with the WGAN synthesized data, respectively. This improvement when training with WGAN synthesized data over the MBGAN synthesized data tracks with the visual differences between the two generation methods. It can also be seen that the best performing CNN and CAE networks were each trained on the SS-GAN synthesized data, with improvements in classification with the CNN of 1% and the CAE of about 2% compared to using the WGAN synthesized data for training. This demonstrates that the proposed SS-GAN synthesized data better represent UAV kinematics than WGAN-GP or MBGAN with envelope extraction.

TABLE VII
DNN Classification Accuracy When Training and Testing Data From Disparate Environments

Network	Training Data	Testing Data	Accuracy
CNN	Real @ High SINR	Real @ Low SINR	41.6%
	SS-GAN — High SINR		28.5%
	Real @ High SINR	Real @ Low SINR w/HERMs	56.1%
	SS-GAN — High SINR		57.7%
CAE	Real @ High SINR	Real @ Low SINR	47.3%
	SS-GAN — High SINR		44.0%
	Real @ High SINR	Real @ Low SINR w/HERMs	50.0%
	SS-GAN — High SINR		71.2%

D. UAV Recognition in Disparate Environments

The purpose of this work is not only to demonstrate a method for better synthesizing kinematically accurate spectrograms of UAV μD, but also to best train a network to classify UAV data in a different environment. As was demonstrated in Section IV, the physical environment matters less than the level of the SINR per sample. Because of this distinction, the CNN and CAE networks from Section VI-C were now tested with the real data at Low SINR for four UAVs (three quadcopters and small hexacopter) because the large hexacopter did not result in any Low-SINR samples. For training the networks, all real High-SINR data for the four UAVs were utilized. This same training dataset was also used to retrain the SS-GAN.

The five-run average classification accuracies for the CNN and CAE are shown in Table VII. The results are

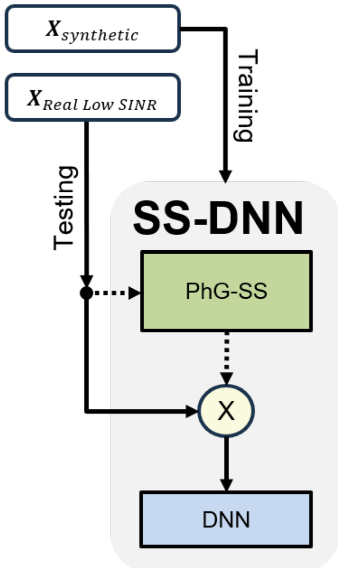


Fig. 15. Semantic segmentation DNN architecture.

also separated into performance when using Low-SINR data exhibiting visible HERM lines (albeit faint) and the entire dataset of Low-SINR data, including samples where HERM lines were not visible. Notice that if only real samples from the disparate (High SINR) environment are utilized for training, neither the CNN nor the CAE is able to achieve an accuracy of even 50% on the entire set of Low-SINR data. In this case, utilization of SS-GAN data for training also does not help—an unsurprising result because SS-GAN is trying to generate HERM lines not often present in all the Low-SINR samples. However, if we look at performance on the subset of Low-SINR data that do exhibit HERM lines, a drastic jump in performance may be seen. The CAE significantly outperforms the CNN as a classifier, and with SS-GAN data for training, just over 71% accuracy can be achieved. This represents a roughly 21% improvement over using real data acquired at high SINR.

VII. SS-DNN: ADDING SEMANTIC SEGMENTATION INTO CLASSIFIER

A. Network Architecture

Further improvements to classification in disparate environments can be achieved by effectively utilizing the PhG-SS network trained with CAD-based simulation as a *denoiser* prior to input to a DNN for classification. As shown in Fig. 15, the resulting SS-DNN first utilizes the PhG-SS network to compute an SM for the input μD signature. The SM is then multiplied with the image itself, essentially functioning as a mask over the data, denoising and suppressing non-UAV related returns. The product is then input into any desired DNN for classification. In this article, we evaluate our results for both the three-layer CNN and the CAE, which each demonstrated inference speeds of 0.068 and 0.0684 s, respectively, when run on Google Colab's T4 GPU.

B. Classification Results

The results of classification with each variant of the SS-DNN are shown in Table VIII. If training with real data from a disparate environment (high SINR), utilization of the semantic segmentation stage with a CNN (SS-CNN) results in a 9% performance improvement. When training on SS-GAN synthesized data, the performance gains are even more dramatic: a 16% jump from 57.7% accuracy to 73.1% accuracy. Performance gains were also seen when utilizing the semantic segmentation stage with a CAE (SS-CAE). In this case, irregardless of the training data utilized, gains of about 2% were observed, resulting in the best classification accuracy of 73.1%.

To further gauge the efficacy of the SS-DNNs, the dataset of real spectrograms at low SINR with distinguishable HERM lines was increased in size by reducing the duration of each μD signature from 6 to 2 s. Thus, the size of the dataset of real samples at high SINR increased to 273 samples for training and 84 samples for testing. Each classification network was tested by randomly selecting 80% of the test samples and repeating the process five times. The results are presented in Table IX, which reports the average accuracy and standard deviation over five iterations. First, we found that comparable results were obtained for both 6- and 2-s duration samples. Second, comparing the results of Tables IV and VI, we found that despite the shorter duration of the data, the benefits of both SS-DNNs for classification offered improvements to recognition accuracy despite the SINR-mismatch of training/testing datasets. Third, utilization of SS-GAN data during pretraining again boosted accuracy relative to that attained from real data.

These results demonstrate the functionality of PhG-SS as a denoiser, suppressing potentially dynamic clutter, as long as distinct HERM lines appear in the μD signature. This is significant because it also shows that utilization of the SS-DNN can improve the ability of classifiers to generalize to different environments, suppressing clutter dynamics. It is also interesting to see that for classification with CNNs, adding the semantic segmentation stage into the classifier resulted in the greatest performance improvements, while for the CAEs, utilization of semantic segmentation in the GAN architecture to improve synthesis resulted in the greater performance improvements. Ultimately, for both networks, the combination of SS-GAN for training data synthesis and SS-DNN for classification resulted in the best overall results—without utilizing any in situ training data consistent with the test environment (SINR). In other words, when classifying UAV signatures at Low SINR, no prior in situ data are utilized. Instead, only simulations and data from a disparate environment (High SINR) were exploited. Consequently, the proposed approach mitigates the performance degradation incurred when seeking to classify targets in different environments, where no in situ training data are available prior to deployment in a potentially dynamic clutter environment with target motion resulting in varying SINR.

TABLE VIII
SS-DNN Classification Accuracy When Training and Testing 6-s Data From Disparate Environments

Network	Training Data	Testing Data	Accuracy
SS-CNN	Real @ High SINR	Real @ Low SINR w/HERMs	65.4%
	SS-GAN — High SINR		73.1%
SS-CAE	Real @ High SINR	Real @ Low SINR w/HERMs	52.6%
	SS-GAN — High SINR		73.1%

TABLE IX
SS-DNN Classification Accuracy for 2-s Samples

Network	Training Data	Testing Data	Accuracy
SS-CNN	Real @ High SINR	Real @ Low SINR w/HERMs	65.38% +/- 0.97%
	SS-GAN — High SINR		67.94% +/- 3.08%
SS-CAE	Real @ High SINR	Real @ Low SINR w/HERMs	65.39% +/- 1.77%
	SS-GAN — High SINR		74.35% +/- 1.72%

VIII. DISCUSSION AND FUTURE WORK

This work proposes a novel way of embedded physics-based knowledge into DNNs: utilization of physics-based simulations to train specific blocks and layers of a network to learn fundamental target properties, in this case, the HERM lines. It is interesting to note that the PhG-SS block offered greater benefits when integrated into the classifier than when utilized to improve the kinematic fidelity of GANs. One view of the functionality of the PhG-SS block in the SS-GAN is that of a denoiser, blocking noisy pixels in regions outside expected HERM lines. It could also be interpreted as a form of physics-based attention, focusing features extracted for classification from the regions where HERM lines are expected. In this way, the SS-GAN exhibits greater robustness to dynamic scenarios where there are variations in SINR that cause training/testing mismatch. Further improvements to the architecture could be achieved by developing improved ways of learning kinematic metrics incorporating them as a physics-based loss in the cost function of the GAN, as in [19].

While the SS-DNN represents an approach exhibiting robustness to dynamic environments, there remain several other aspects impacting classification accuracy that remain challenges. First, this work focuses on the proposed techniques as applied to radar μD signatures, whereas the radar actually obtains additional measurements that could be exploited for classification, such as range–Doppler and range–angle measurements. Improvements to the DNNs utilized in this work could be made by incorporating joint domain multi-input multitask learning [62] to capture additional features. For example, this could involve varying the window lengths of the STFT to potentially capture blade flashes and/or HERM lines, learning optimal time–frequency transformations instead of hard-coding STFT parameters, incorporating range information that captures the flight track of the UAV, and learning matched filters for improved detection in low SINR scenarios.

In addition, further robustness to dynamic operational conditions could be attained by incorporating more realistic dynamic clutter models and RF interference scenarios. One way in which more complex sources of clutter can be accounted for by the model while still being invariant across environments would be to expand the complexity of clutter models injected into the simulated dataset used to train the PhG-SS.

Additional studies on HERM line representation in radar data can also be pursued through the use of higher dimensional 3-D data representations and the exploitation of video-synthesis methods, such as TempoGAN [49]. While TempoGAN uses the temporal information of dynamic processes inherent in a series of images of said process, future work on capturing the temporal features of UAV μD HERM lines will require a method of input that accounts for the temporal variation within a single spectrogram, not across multiple spectrograms. More generally, the question of how to best process and apply machine learning techniques to 4-D or 5-D radar tensors for ATR in real time for real-world operational scenarios remains a challenging open problem.

IX. CONCLUSION

This article addressed the twofold challenge of 1) how to accurately synthesize HERM lines in training data for UAV classification and 2) how to enhance recognition performance in previously unseen different environments, especially when the SINR is low. Toward these aims, we developed physics-guided DNNs for UAV μD signature synthesis and classification. The HERM lines present in UAV μD signatures represent unique features for UAV recognition, but they are often erroneously represented in samples synthesized by GANs. Our proposed approach, which integrates a PhG-SS network trained on CAD-model-based simulations into an auxiliary branch of the discriminator, was shown to maintain kinematically accurate HERM lines in the synthetic samples, thereby boosting classification accuracy. In addition, we also proposed an SS-DNN classifier, which utilizes the PhG-SS network as a mask to suppress nontarget returns and essentially denoise the data prior to classification. Our results showed that utilization of the proposed SS-GAN for training data generation and SS-DNN for classification of μD signatures exhibiting HERM lines results in 23% improvement in classification accuracy—without using any prior in situ data for training. Consequently, our approach provides a way of enhancing UAV recognition performance in different operational environments, from which it is often difficult to obtain training data prior to deployment.

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REFERENCES

[1] S. Z. Gurbuz, *Deep Neural Network Design for Radar Applications*. London, U.K.: IET, 2020.

[2] *Deep Learning for Radar and Communications Automatic Target Recognition*. Boston, MA, USA: Artech House, 2020.

[3] F. Hoffmann, M. Ritchie, F. Fioranelli, A. Charlish, and H. Griffiths, "Micro-Doppler based detection and tracking of UAVs with multistatic radar," in *Proc. IEEE Radar Conf. (RadarConf)*, 2016, pp. 1–6.

[4] N. Rojhani, M. Passafiume, M. Sadeghibakhi, G. Collodi, and A. Cidronali, "Model-based data augmentation applied to deep learning networks for classification of micro-Doppler signatures using FMCW radar," *IEEE Trans. Microw. Theory Techn.*, vol. 71, no. 5, pp. 2222–2236, May 2023.

[5] B. I. Ahmad et al., "A review of automatic classification of drones using radar: Key considerations, performance evaluation, and prospects," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 39, no. 2, pp. 18–33, Feb. 2024.

[6] D. White, M. Jahangir, A. Kumar Mishra, C. J. Baker, and M. Antoniou, "Latent variable and classification performance analysis of bird–drone spectrograms with elementary autoencoder," *IEEE Trans. Radar Syst.*, vol. 3, pp. 115–123, 2025.

[7] F. Fioranelli, M. Ritchie, H. Griffiths, and H. Borrion, "Classification of loaded/unloaded micro-drones using multistatic radar," *Electron. Lett.*, vol. 51, no. 22, pp. 1813–1815, 2015.

[8] M. Ritchie, F. Fioranelli, H. Borrion, and H. Griffiths, "Multistatic micro-Doppler radar feature extraction for classification of unloaded/loaded micro-drones," *IET Radar, Sonar Navig.*, vol. 11, no. 1, pp. 116–124, 2017.

[9] M. Ritchie, F. Fioranelli, H. Griffiths, and B. Torvik, "Monostatic and bistatic radar measurements of birds and micro-drone," in *Proc. IEEE Radar Conf.*, 2016, pp. 1–5.

[10] M. Jahangir, G. Atkinson, M. Antoniou, C. Baker, J. Sadler, and S. Reynolds, "Measurements of birds and drones with L-band staring radar," in *Proc. 21st Int. Radar Symp.*, 2021, pp. 1–10.

[11] R. Palamà, F. Fioranelli, M. Ritchie, M. Inggs, S. Lewis, and H. Griffiths, "Measurements and discrimination of drones and birds with a multi-frequency multistatic radar system," *IET Radar, Sonar Navig.*, vol. 15, no. 8, pp. 841–852, 2021.

[12] R. M. Narayanan, B. Tsang, and R. Bharadwaj, "Classification and discrimination of birds and small drones using radar micro-Doppler spectrogram images," *Signals*, vol. 4, no. 2, pp. 337–358, 2023.

[13] A. Pappas, J. J. De Wit, F. Fioranelli, and B. Jacobs, "Multitask learning for radar-based characterization of drones," in *Proc. IEEE Int. Radar Conf.*, 2023, pp. 1–7.

[14] S. Z. Gurbuz, M. M. Rahman, E. Kurtoğlu, T. Macks, and F. Fioranelli, "Cross-frequency training with adversarial learning for radar micro-Doppler signature classification (rising researcher)," *Proc. SPIE*, vol. 11408, pp. 58–68, 2020.

[15] I. J. Goodfellow et al., "Generative adversarial nets," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.

[16] M. Mirza and S. Osindero, "Conditional generative adversarial nets," 2014, *arXiv:abs/1411.1784*.

[17] A. Radford, L. Metz, and S. Chintala, "Unsupervised representation learning with deep convolutional generative adversarial networks," 2016, *arXiv:abs/1511.06434*.

[18] B. Erol, S. Z. Gurbuz, and M. G. Amin, "Motion classification using kinematically sifted ACGAN-synthesized radar micro-Doppler signatures," *IEEE Trans. Aerosp. Electron. Sys.*, vol. 56, no. 4, pp. 3197–3213, Aug. 2020.

[19] M. Rahman, S. Gurbuz, and M. Amin, "Physics-aware generative adversarial networks for radar-based human activity recognition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 3, pp. 2994–3008, Jun. 2023.

[20] M. M. Rahman, E. A. Malaia, A. C. Gurbuz, D. J. Griffin, C. Crawford, and S. Z. Gurbuz, "Effect of kinematics and fluency in adversarial synthetic data generation for ASL recognition with RF sensors," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 4, pp. 2732–2745, Aug. 2022.

[21] S. Rahman and D. A. Robertson, "Millimeter-wave micro-Doppler measurements of small UAVs," *Proc. SPIE*, vol. 10188, 2017, Art. no. 101880T.

[22] B. Choi and D. Oh, "Classification of drone type using deep convolutional neural networks based on micro-Doppler simulation," in *Proc. Int. Symp. Antennas Propag.*, 2018, pp. 1–2.

[23] C. Bennett, S. Harman, and I. Petrunin, "Realistic simulation of drone micro-Doppler signatures," in *Proc. 18th Eur. Radar Conf.*, 2022, pp. 114–117.

[24] M. Moore, D. A. Robertson, and S. Rahman, "Simulating UAV micro-Doppler using dynamic point clouds," in *Proc. IEEE Radar Conf.*, 2022, pp. 1–6.

[25] Y. Cai, O. Krasnov, and A. Yarovoy, "Simulation of radar micro-Doppler patterns for multi-propeller drones," in *Proc. Int. Radar Conf.*, 2019, pp. 1–5.

[26] F. Fioranelli, O. Krasnov, Y. Cai, A. Yarovoy, J. Yun, and D. Anderson, "Improving the simulations of radar signatures of small drone," *MSG-SET-183 Specialists' Meeting Drone Detectability: Modelling Relevant Signature*, NATO Science and Technology Organization (STO), Apr., 2021.

[27] D. White et al., "Multi-rotor drone micro-Doppler simulation incorporating genuine motor speeds and validation with L-band staring radar," in *Proc. IEEE Radar Conf.*, 2022, pp. 1–6.

[28] D. White, M. Jahangir, C. J. Baker, and M. Antoniou, "Urban bird-drone classification with synthetic micro-Doppler spectrograms," *IEEE Trans. Radar Syst.*, vol. 2, pp. 167–179, 2024.

[29] H. Dale, C. Baker, M. Antoniou, M. Jahangir, and G. Atkinson, "A comparison of convolutional neural networks for low SNR radar classification of drones," in *Proc. IEEE Radar Conf.*, 2021, pp. 1–5.

[30] H. Dale, M. Jahangir, C. J. Baker, M. Antoniou, S. Harman, and B. I. Ahmad, "Convolutional neural networks for robust classification of drones," in *Proc. IEEE Radar Conf.*, 2022, pp. 1–6.

[31] E. P. Blasch, S. Ravela, and A. J. Aved, "The dynamic data driven applications systems (DDDAS) paradigm and emerging directions," in *Handbook of Dynamic Data Driven Applications Systems*, vol. 2. New York, NY, USA: Springer, 2023.

[32] S. Z. Gurbuz, "Physics-aware machine learning for dynamic, data-driven radar target recognition," in *Proc. Dyn. Data-Driven Appl. Syst. Conf.*, Oct. 2022, pp. 114–122.

[33] P. Kaul, D. de Martini, M. Gadd, and P. Newman, "RSS-Net: Weakly-supervised multi-class semantic segmentation with FMCW radar," in *Proc. IEEE Intell. Veh. Symp.*, 2020, pp. 431–436.

[34] J. Rebut, A. Ouaknine, W. Malik, and P. Pérez, "Raw high-definition radar for multi-task learning," in *Proc. IEEE/CVF Conf. Comput. Vis/ Pattern Recognition (CVPR)*, 2022, pp. 17021–17030.

[35] L. Zhang et al., "PeakConv: Learning peak receptive field for radar semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Los Alamitos, CA, USA, Jun. 2023, pp. 17577–17586.

[36] Y. Dalbah, J. Lahoud, and H. Cholakkal, "TransRadar: Adaptive-directional transformer for real-time multi-view radar semantic segmentation," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, 2023,pp. 353–362.

[37] A. N. Sayed, O. M. Ramahi, and G. Shaker, "Detection and classification of drones using radars, AI, and full-wave electromagnetic cad tool," in *Drones—Various Applications*, D. Cvetković, Ed. Rijeka, Croatia: IntechOpen, 2023, ch. 2.

[38] B. Erol, S. Z. Gurbuz, and M. G. Amin, "GAN-based synthetic radar micro-Doppler augmentations for improved human activity recognition," in *Proc. Proc. IEEE Radar Conf.*, 2019, pp. 1–5.

[39] I. Alnujaim, D. Oh, and Y. Kim, "Generative adversarial networks for classification of micro-Doppler signatures of human activity," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 3, pp. 396–400, Mar. 2020.

[40] H. G. Doherty, L. Cifola, R. I. A. Harmanny, and F. Fioranelli, "Unsupervised learning using generative adversarial networks on micro-Doppler spectrograms," in *Proc. 16th Eur. Radar Conf.*, 2019, pp. 197–200.

[41] M. L. L. de Oliveira and M. J. G. Bekooij, "Generating synthetic short-range FMCW range-Doppler maps using generative adversarial networks and deep convolutional autoencoders," in *Proc. IEEE Radar Conf.*, 2020, pp. 1–6.

[42] J. Liu, R. Ewing, E. Blasch, and J. Li, "Synthesis of passive human radio frequency signatures via generative adversarial network," in *Proc. IEEE Aerosp. Conf.*, 2021, pp. 1–9.

[43] J. Long, A. Hiromori, and A. Uchiyama, "A preliminary study on synthesizing mmWave radar range-Doppler maps under occlusion," in *Proc. IEEE Int. Conf. Pervasive Comput. Commun. Workshops*, 2025, pp. 355–360.

[44] Y. Lang, C. Hou, H. Ji, and Y. Yang, "A dual generation adversarial network for human motion detection using micro-Doppler signatures," *IEEE Sensors J.*, vol. 21, no. 16, pp. 17995–18003, Aug. 2021.

[45] L. Qu, Y. Wang, T. Yang, and Y. Sun, "Human activity recognition based on WRGAN-GP-synthesized micro-Doppler spectrograms," *IEEE Sensors J.*, vol. 22, no. 9, pp. 8960–8973, May 2022.

[46] Y. Yang, Y. Zhang, H. Ji, B. Li, and C. Song, "Radar-based human activity recognition under the limited measurement data support using domain translation," *IEEE Signal Process. Lett.*, vol. 29, pp. 1993–1997, 2022.

[47] I. Alnujaim, S. S. Ram, D. Oh, and Y. Kim, "Synthesis of micro-Doppler signatures of human activities from different aspect angles using generative adversarial networks," *IEEE Access*, vol. 9, pp. 46422–46429, 2021.

[48] H. Li, Y. Xiang, H. Xu, and W. Wang, "Pedestrian recognition with radar data-enhanced deep learning approach based on micro-Doppler signatures," in *Proc. IEEE 35th Int. Conf. Tools Artif. Intell.*, Los Alamitos, CA, USA, Nov. 2023, pp. 437–443.

[49] Y. Xie, A. Franz, M. Chu, and N. Thuerey, "tempoGAN: A temporally coherent, volumetric GAN for super-resolution fluid flow," *ACM Trans. Graph.*, vol. 37, no. 4, Aug. 2018, Art. no. 95.

[50] S. Park, S. Lee, and N. Kwak, "Range-Doppler map augmentation by generative adversarial network for deep UAV classification," in *Proc. IEEE Radar Conf.*, 2022, pp. 1–7.

[51] O. Schumann, M. Hahn, J. Dickmann, and C. Wöhler, "Semantic segmentation on radar point clouds," in *Proc. 21st Int. Conf. Inf. Fusion*, 2018, pp. 2179–2186.

[52] A. Ouaknine, A. Newson, P. Pérez, F. Tupin, and J. Rebut, "Multi-view radar semantic segmentation," in *Proc. IEEE/CVF Int. Conference Comput. Vis. (ICCV)*, 2021, pp. 15671–15680.

[53] F. Ahmad, M. G. Amin, and G. Mandapati, "Autofocusing of through-the-wall radar imagery under unknown wall characteristics," *IEEE Trans. Image Process.*, vol. 16, no. 7, pp. 1785–1795, Jul. 2007.

[54] M. Zenaldin and R. M. Narayanan, "Radar micro-Doppler based human activity classification for indoor and outdoor environments," *Proc. SPIE*, vol. 9829, 2016, Art. no. 98291B.

[55] J. Wang, T. Zheng, P. Lei, and X. Bai, "Ground target classification in noisy SAR images using convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 4180–4192, Nov. 2018.

[56] N. Inkawhich et al., "Bridging a gap in SAR-ATR: Training on fully synthetic and testing on measured data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2942–2955, 2021.

[57] H. Dale, C. Baker, M. Antoniou, M. Jahangir, G. Atkinson, and S. Harman, "SNR-dependent drone classification using convolutional neural networks," *IET Radar, Sonar Navig.*, vol. 16, no. 1, pp. 22–33, 2022.

[58] E. Ozturk, F. Erden, and I. Guvenc, "RF-based low-SNR classification of UAVs using convolutional neural networks," *ITU J. Future Evolving Technol.*, vol. 2, pp. 39–52, 2021.

[59] V. Chen, *The Micro-Doppler Effect in Radar,* 2nd ed. London, U.K.: Artech House, 2019.

[60] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," *Med. Image Comput. Comput.-Assist. Interv.*, vol. 9351, Navab, N., Hornegger, J., Wells, W., Frangi, A. eds., Springer, Cham, 2015. https://doi.org/10.1007/978-3-319-24574-4_28

[61] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, Jun. 2002.

[62] E. Kurtoğlu, A. C. Gurbuz, E. A. Malaia, D. Griffin, C. Crawford, and S. Z. Gurbuz, "ASL trigger recognition in mixed activity/signing sequences for RF sensor-based user interfaces," *IEEE Trans. Human-Mach. Syst.*, vol. 52, no. 4, pp. 699–712, Aug. 2022.

Sean Kearney (Student Member, IEEE) received the B.S. and M.S. degrees in electrical engineering from the University of Alabama, Tuscaloosa, AL, USA, in 2021 and 2024, respectively. He is currently working toward the Ph.D. degree in electrical and computer engineering with the Lab of Computational Imaging for Radar, North Carolina State University, Raleigh, NC, USA.

In 2021 and 2022, he was an intern with the U.S. Naval Research Lab. In 2025, he was an intern with the U.S. Air Force Research Lab Sensors Directorate. His research interests include adaptive target recognition, radar signal processing, and physics-aware machine learning.

Mr. Kearney was the recipient of first place in the 2025 AERPAW Community Workshop Poster Competition, second place in the 2025 IEEE International Radar Conference Student Paper Competition, third place in 2024 IEEE Radar Conference Radar Challenge, an honorable citation award at 2024 DDDAS/Infosymbiotics Conference, and a Fellowship by the Alabama EPSCoR Graduate Research Scholars Program in Fall 2024.

Sevgi Z. Gurbuz (Senior Member, IEEE) received the B.S. degree in electrical engineering with minor in mechanical engineering and the M.Eng. degree in electrical engineering and computer science from the Massachusetts Institute of Technology, Cambridge, MA, USA, in 1998 and 2000, respectively, and the Ph.D. degree in electrical and computer engineering from the Georgia Institute of Technology, Atlanta, GA, USA, in 2009.

From 2000 to 2004, she was a Radar Signal Processing Research Engineer with the U.S. Air Force Research Laboratory, Sensors Directorate, Rome, NY, USA. Subsequently, she was a Senior Research Scientist with the TUBITAK Space Technologies Research Institute, Ankara, Türkiye; an Assistant Professor of Electrical-Electronics Engineering with the TOBB University of Economics and Technology, Ankara; and an Associate Professor of Electrical and Computer Engineering with the University of Alabama, Tuscaloosa, AL, USA. Since 2025, she has been an Associate Professor of Electrical and Computer Engineering with North Carolina State University, Raleigh, NC, USA, and the Director of the Lab of Computational Imaging for Radar. Her current research interests include RF sensing, radar signal processing, physics-aware machine learning, and RF-enabled cyber-physical human systems for applications of human motion recognition for remote health monitoring and human–computer interaction applications.

Dr. Gurbuz is a recipient of a 2023 NSF CAREER Award, the IEEE Harry Rowe Mimno Award for 2019, the 2020 SPIE Rising Researcher Award, EU Marie Curie Research Fellowship, and the 2010 IEEE Radar Conference Best Student Paper Award. She is also a Senior Member of the SPIE, and a Member of ACM and Eta Kappa Nu.